

PROBLEM IDENTIFICATION AND FEASIBILITY ANALYSIS

QR-Based Restaurant Ordering System

1. Problem Identification

Background. In a traditional restaurant, customers generally depend on restaurant staff to provide menus, take food orders, and communicate those orders to the kitchen. During busy periods, this manual process can increase waiting time and may result in mistakes while recording or communicating orders.

Identified Problem. The selected software project addresses the difficulty of managing restaurant orders efficiently through a completely manual process. Customers may have to wait for staff to become available before ordering, while staff members have to manually record and communicate multiple orders. This can lead to delays, communication errors, and inefficient order management.

Proposed Solution. The QR-Based Restaurant Ordering System provides a digital alternative. A QR code is placed on each restaurant table. Customers scan the QR code using their smartphones, access the digital menu, select food items, add them to a cart, and place the order. The order is then processed through the backend and stored in the database so that restaurant staff can manage it digitally.

2. Problem Statement

Traditional restaurant ordering systems rely heavily on manual interaction between customers and staff, which can cause delays, communication errors, and inefficient order management, particularly during busy periods. There is therefore a need for a simple and reliable digital ordering system that allows customers to place orders directly from their tables while enabling restaurant staff to receive and manage those orders efficiently.

3. Objectives

- To provide a fast and contactless method for placing restaurant orders.
- To reduce errors associated with manually taking and recording customer orders.
- To improve the overall customer ordering experience.
- To allow restaurant staff to manage incoming orders digitally.
- To reduce unnecessary waiting time during the ordering process.
- To maintain restaurant order information in a structured digital form.
- To demonstrate the practical application of software engineering and design principles.

4. Scope of the Project

4.1 Customer Scope

- Scanning the QR code assigned to a restaurant table.
- Accessing and viewing the digital menu.
- Selecting available food items.
- Adding and removing items from the cart.
- Placing an order through the system.
- Viewing the status of an order.

4.2 Restaurant Staff Scope

- Receiving customer orders through the digital system.
- Viewing and managing incoming orders.
- Updating the status of orders during preparation.
- Marking orders as completed.

4.3 System Scope

The system covers the flow from table identification through QR code scanning to digital menu access, cart management, order placement, backend processing, database storage, and restaurant-side order management.

4.4 Limitations and Future Scope

The current project focuses on the core restaurant ordering process. Features such as online payment, notifications, and support for multiple restaurants can be added as future extensions.

5. Feasibility Analysis

Feasibility analysis determines whether the proposed software project can be successfully developed and used in a practical environment. For the QR-Based Restaurant Ordering System, technical and operational feasibility are considered.

5.1 Technical Feasibility

The project is technically feasible because it can be developed using commonly available web technologies and development tools. The system uses HTML and CSS for the user interface, JavaScript for frontend functionality, Node.js and Express.js for backend/API operations, and MySQL for database storage. QR codes are used for table identification, while Git and GitHub support version control and collaboration.

The system follows a three-layer architecture consisting of the Customer UI, Backend/API, and Database. This separation keeps the user interface, business logic, and data storage organized into distinct layers. The required software technologies are widely available, and customers can access the system using their smartphones.

- Required web development technologies are available and suitable for the project.

- A smartphone can be used by customers to scan the table QR code and access the menu.
- Node.js and Express.js can provide the required backend/API functionality.
- MySQL can store menu, customer order, and order-status information.
- The modular architecture allows the system to be tested and extended.

Technical Feasibility Result: The project is technically feasible with the proposed technologies and architecture.

5.2 Operational Feasibility

The project is operationally feasible because its workflow is simple and does not require customers or restaurant staff to learn a complicated procedure. Customers only need to scan the QR code, view the menu, select items, add them to the cart, and place the order. Restaurant staff can receive and manage the orders through the restaurant-side interface.

The system can improve daily restaurant operations by reducing dependency on manual order-taking, reducing communication errors, and making order information available digitally. The digital workflow can also help staff manage incoming orders more systematically.

- The customer workflow is simple and easy to understand.
- No special ordering device is required for customers beyond a smartphone.
- Restaurant staff can manage orders digitally.
- The system supports a faster and more organized ordering process.
- The proposed workflow can be integrated into normal restaurant operations.

Operational Feasibility Result: The project is operationally feasible for use in a restaurant environment.

6. Conclusion

The QR-Based Restaurant Ordering System addresses the problems associated with manual restaurant ordering by providing a simple digital ordering workflow. The system allows customers to order directly from their tables using QR codes and enables restaurant staff to manage orders digitally.

Based on the technical and operational analysis, the proposed system is feasible to develop and operate. The selected technologies are appropriate for implementing the system, and the workflow is simple enough to be adopted by customers and restaurant staff. Therefore, the project is a practical software solution for improving the efficiency of restaurant ordering.